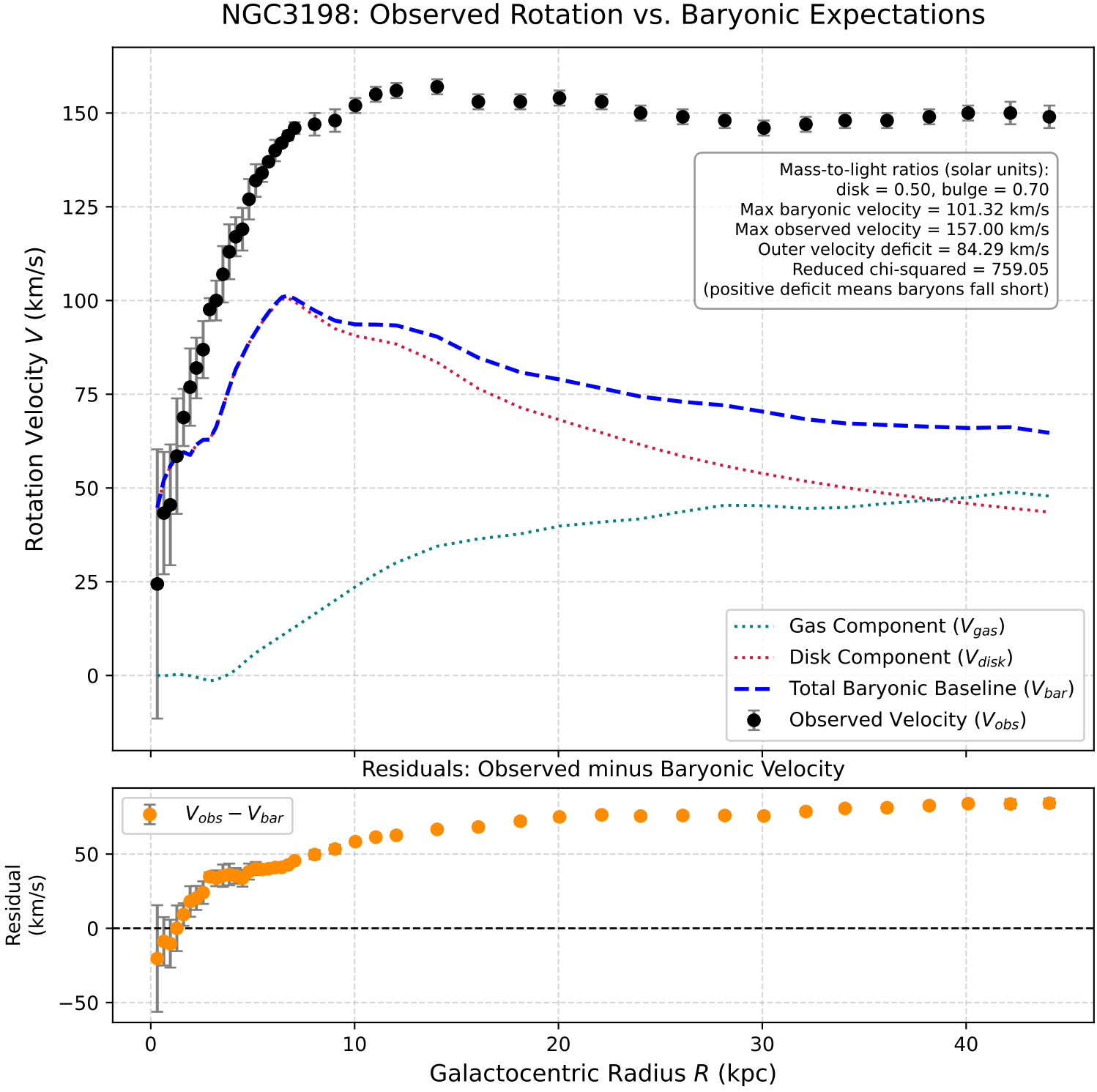
NGC3198: Observed Rotation vs. Baryonic Expectations
Rotation Velocity V (km/s)
Mass-to-light ratios (solar units):
disk = 0.50, bulge = 0.70
Max baryonic velocity = 101.32 km/s
Max observed velocity = 157.00 km/s
Outer velocity deficit = 84.29 km/s
Reduced chi-squared = 759.05
(positive deficit means baryons fall short)
Gas Component (V_{gas})
Disk Component (V_{disk})
Total Baryonic Baseline (V_{bar})
Observed Velocity (V_{obs})
Residuals: Observed minus Baryonic Velocity
$V_{obs} - V_{bar}$
Residual (km/s)
Galactocentric Radius R (kpc)